Shane Patandin

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Education

Stony Brook University — Stony Brook, NY
Master of Science: Data Science — *Expected December 2026*

Stony Brook University — Stony Brook, NY
Bachelor of Science: Applied Mathematics and Statistics, Economics — GPA: 3.5 — *May 2025*

Relevant Coursework: Statistical Learning, Big Data Systems, Data Management, Data Mining, Machine Learning in Quantitative Finance, Mathematical Statistics, Econometrics

Experience

Artemis Speakeasy — Ronkonkoma, NY
Data Scientist Intern — *May 2026 – Present*

- Engineered a PostgreSQL data warehouse for 17 months of restaurant sales data (about $2.3M), with automated testing and a 16x faster refresh (96s to 6s).
- Corrected an initial $156K/yr estimate for faster table turns to about $9K by showing that revenue was constrained by demand rather than turnover, with 29% average occupancy.
- Quantified an annual decline of about 14% in customer volume after adjusting for seasonality, offset in part by an 8% rise in spend per guest driven by menu mix.

Stony Brook University — Stony Brook, NY
Research Assistant — *June 2026 – Present*

- Engineering MetaGlass, an Android application that helps blind users locate and interpret visual charts/graphs in real time using Meta Ray-Ban smart glasses, combining computer vision, sensor fusion, and audio feedback.
- Building a real-time multi-phase pipeline (scan, align, capture, analyze, narrate) that integrates ML Kit object detection and text recognition with a hybrid spatial-tracking system (about 5–15ms latency) to guide users via directional audio cues.
- Integrating the lab's LLM chart interpretation system for multimodal chart analysis, using OpenCV pre-cropping and bounding box refinement to improve description accuracy, with results narrated through text to speech.

Stony Brook University — Stony Brook, NY
Resident Assistant (RA) — *August 2024 – Present*

- Mentor and support 100+ residents as a first point of contact through on call duty, front desk check ins, and community programming for students from many countries.
- Surveyed 109 RAs with a partner and applied NLP to their written responses, presenting recommendations on programming requirements to the university's AICE committee.

Projects

Traffic Safety Analysis on Long Island | GitHub — April 2026

- Identified four accident archetypes across 33,000+ Long Island crash records using FAMD dimensionality reduction with k-Means, reaching a silhouette score of 0.43 versus 0.16 for a k-Prototypes baseline.
- Built an XGBoost severity classifier on 26 engineered weather, infrastructure, and temporal features, with 5-fold CV confirming stable out of sample performance.
- Detected 12 accident corridors along the LIE and Northern/Southern State Parkways with DBSCAN on projected coordinates, cross-referenced against the archetypes to target safety interventions.

Skills

Programming Languages: Python, R, SQL, SAS
Methods: Machine Learning, Regression, Classification, Clustering, NLP, Time Series, Statistical Modeling
Libraries: Scikit-learn, XGBoost, LightGBM, Optuna, PyTorch, OpenCV, Pandas, NumPy, Matplotlib, ggplot2
Databases & Tools: PostgreSQL, SQLAlchemy, Docker, Git, GitHub, Android Studio